Stage2 Gears — ISO 6336 Bending & Contact Stress Check

Tooth Counts & Ratio Check		
Pinion tooth count, z1	25	
Gear tooth count, z2	40	
Gear ratio, u = z2/z1	1.60	
Target stage ratio (from Inputs)	1.58	
Geometry		
Normal module, mn	3.00	mm
Face width, b (trial - iterate to hit target SF)	30.0	mm
Pinion pitch diameter, d1 = mn*z1/cos(beta)	75.00	mm
Gear pitch diameter, d2 = mn*z2/cos(beta)	120.00	mm
Center distance, a = (d1+d2)/2	97.50	mm
Face width ratio, b/d1	0.40	
Loads		
Input speed at this pinion	7278.5	rpm
Input torque at this pinion	94.6	N·m
Pitch line velocity, v = pi*d1*n/60000	28.58	m/s
Tangential load, Ft = 2000*T/d1	2521.7	N
Transverse Contact Ratio — CALCULATED (real involute geometry)		
Pinion base radius, rb1 = (mn*z1/2)*cos(alpha_n)	35.24	
Gear base radius, rb2 = (mn*z2/2)*cos(alpha_n)	56.38	
Pinion tip radius, ra1 = mn*z1/2 + mn (std addendum)	40.50	
Gear tip radius, ra2 = mn*z2/2 + mn (std addendum)	63.00	
Transverse contact ratio, eps_alpha	1.663	
Load & Application Factors		
Application factor, KA (guided input - see README)	1.50	
Dynamic factor, Kv — CALCULATED (AGMA-equivalent closed form, see README)		
Quality number, Qv (from Inputs)	6.0	
B = 0.25*(12-Qv)^(2/3)	0.825	
A = 50 + 56*(1-B)	59.77	
v in m/min = v(m/s)*60	1715.0	
Kv = ((A+SQRT(3.28*v_m/min))/A)^B	1.956	
Face load factor - contact, KHb (guided - see README, b/d1 sh	1.25	
Face load factor - bending, KFb (guided, ~ KHb but slightly lowe	1.22	
Transverse load factor - contact, KHa (guided, Grade 8 typical)	1.10	
Transverse load factor - bending, KFa (guided, Grade 8 typical)	1.10	
ISO 6336-3 Bending Geometry Factors		
Contact ratio factor, Yeps = 0.25 + 0.75/eps_alpha — CALCULA	0.701	
Helix angle factor, Ybeta = 1 - eps_beta*(beta/120) — CALCULA	1.000	
Tooth form factor, YFa (guided - chart value for z~24, see READ	2.70	

Stress correction factor, YSa (guided - chart value for z~24)	1.60	
ISO 6336-2 Contact Geometry Factors		
Zone factor, ZH — CALCULATED	2.495	
Elasticity factor, ZE (steel-steel, fixed) — CALCULATED (consta	189.8	
Contact ratio factor, Zeps = SQRT((4-eps_alpha)/3) — CALCUL	0.883	
Helix angle factor, Zbeta = SQRT(cos(beta)) — CALCULATED (	1.000	
Material & Life Factors		
Bending endurance limit, sigma_Flim (from Inputs)	430	
Contact endurance limit, sigma_Hlim (from Inputs)	1500	
Stress correction factor, YST (standard reference gear)	2.00	
Life factor - bending, YNT (1.0 = finite/reference life; >1 if very lo	1.10	
Life factor - contact, ZNT	1.10	
Other bending factors lumped, YdeltaT*YRrelT*YX	1.00	
Other contact factors lumped, ZL*ZV*ZR*ZW*ZX	1.00	
Minimum required bending safety factor, SFmin	1.00	
Minimum required contact safety factor, SHmin	1.00	
Bending Stress Check (ISO 6336-3)		
sigma_F = Ft*KA*Kv*KFb*KFa*YFa*YSa*Ybeta*Yeps/(b*mn)	334.2	MPa
sigma_FP = sigma_Flim*YST*YNT*Yother / SFmin	946.0	MPa
Bending safety factor, SF = sigma_FP / sigma_F	2.83	
Contact Stress Check (ISO 6336-2)		
sigma_H0 = ZE*ZH*Zeps*Zbeta*SQRT(Ft*(u+1)/(b*d1*u))	564.0	MPa
sigma_H = sigma_H0 * SQRT(KA*Kv*KHb*KHa)	1133.0	MPa
sigma_HP = sigma_Hlim*ZNT*Zother / SHmin	1650.0	MPa
Contact safety factor, SH = sigma_HP / sigma_H	1.46	